

High-Level Design Models

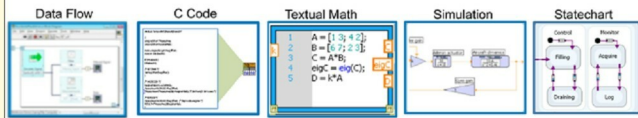


Fig. 1: Multiple methods (models of computation) for developing systems

able off-the-shelf options to explore solutions. The common hardware platform elements—processing (processors, DSP and FPGA), communications and modular input/output—are abstracted at a system level in the same way as abstracting models and other software constructs. Once engineers invest in the learning curve of a system platform, they can integrate and iterate quickly at each stage of the product development cycle.

At the extreme edge of this problem is next-generation development in aerospace and defence. Recently, Defense Advanced Research Projects Agency (DARPA) of USA announced funding research for the META programme, which seeks to improve and accelerate development of defence systems that incorporate software and electromechanical systems through model-based design methods. The graphical system design approach scales much better with these requirements than traditional methods because it abstracts the complexity of system components—both software and hardware—and focuses on the integration of those components.

The productivity benefits of the graphical system design platform span every industry in which engineers create systems that need measurement and control. At the Institute for Plasma Research, Gandhinagar, for example, engineers use graphical system design to monitor temperature, pressure and flow of water in their 82.6GHz, 200kW Gyrotron vacuum tube used for radio frequency (RF) heating. The platform approach to system design provided them with a single learning curve for software

and hardware, reducing their development time from a few months to weeks.

Graphical system design scales to solve automated measurement systems where the software platform abstracts the elements of the input/output, analysis, communications and visualisation needed to upgrade or integrate instrumentation functionality. Texas Instruments, for example, used graphical system design for their automated test platform and reduced test time by 70 per cent, while increasing test coverage by more than 100 per cent for production testing of power management ICs.

Graphical system design is a reconfigurable system approach for developing systems that need measurement and control. Using this approach, engineers can invest in a single development platform from which multiple solutions can be derived to serve a broad range of application needs.

Scientists at the Indian Institute of Technology (IIT), Kharagpur, discovered this benefit while developing a vision-based electronic navigation system for the visually impaired. The system helps to analyse the surroundings in detail. The initial performance trade-off expected by using a customisable off-the-shelf platform as opposed to custom design never materialised. In fact, they exceeded their performance requirements while significantly reducing development and prototyping time.

Platform ecosystem drives innovation

Engineers can leverage the work of other engineers using the graphical system

design approach. One of the benefits of platform-based development is the ecosystem of IP and applications that are regularly shared within the community. In fact, the adoption of virtual instrumentation in the test industry, which was supported by the ecosystem of PCs in software and input/output peripherals, is one example of how the graphical system design approach revolutionised the ability for engineers to gain significantly higher performance at lower costs for a specific application space faster.

Many engineering tools and approaches focus on optimising specific areas of the design to implementation process, even while painting the picture of an integrated flow. Often these are platforms where either software or hardware takes the primary focus and the integration of the overall system is de-prioritised. Engineers can easily select tool chains that continue to require more expertise and time to get to the real solution.

With graphical system design, engineers gain an open reconfigurable platform that accesses multiple technologies and tools at a high level, shortening the most time-consuming portion of the design process. It also generates an ecosystem of IP so that an individual or small team of engineers can leverage years of development without investing additional resources. While many engineers have already had considerable success with this method, graphical system design remains an accessible opportunity for many looking to gain disruptive competitive advantage. ●